Research Notes on ζ(9): Constructions, Computations, and Open Problems

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Status statement (read first). This note does not claim a proof of the irrationality of ζ(9). No such proof exists in the material described here, conditionally or otherwise. The purpose of this version (v0.1) is to make public the concrete constructions, the local arguments that *are* proved (in writing, with the caveats stated), the computations that were actually run and archived, and the open problems that remain. Every substantial statement carries a status tag: **R** a written proof exists and no error was found on review; **R*** proved conditional on a named standard input (e.g. the prime number theorem); **M** proved only modulo another claim in the same corpus; **N** verified numerically (in places with interval arithmetic) on an explicitly finite range, with no uniformity; **H** heuristic, not a proof step; **OPEN** needed but absent. "Not formalized" is reported as a fact about the state of formal verification, *not* as a logical gap in a written proof; conversely, no statement is claimed to be formally verified unless it is.

1. Aim, scope, and provenance

These notes record a two-day research effort (24–25 September 2026) on the irrationality of ζ(9), carried out in a local workspace and partially mirrored into a private mission on a crowdsourced formalization platform. The effort comprises **nine research rounds** (round 1 and rounds 2–9), a **roadmap dependency graph** (40 named nodes) that redirected the programme after round 9, and a **platform layer** containing eight theorem-level formal proofs with *abstract* statements. A full proof-state audit of the corpus was prepared on 25 September 2026 and is included in the source archive as provenance; the present note is the public-facing distillation of that material, with three factual corrections applied (Section 8).

What is published here. (i) Two concrete families of constructions producing rational linear forms in ζ(9) (Part I); (ii) the exact algebra, integrality theorems, valuation formulas and a certified analytic *upper* bound around them; (iii) finite certified computations with archived artifacts and reproduction instructions; (iv) the later constructions and the current roadmap with its open nodes (Part II); (v) a canonical list of what is missing.

What is not claimed. No irrationality statement; no eventual nonvanishing of any linear form along an infinite sequence; no decay of any integer primitive form; no asymptotic for any true primitive denominator; no formal verification of the note-proved layer. Finite computations are reported as finite computations: a value certified at a tested parameter is never presented as a statement about infinitely many parameters, and the first success inside a discrete scan is never presented as a minimal threshold or a final answer.

2. Conventions

Rising Pochhammer $(x)_r = x(x+1)\cdots(x+r-1)$. Harmonic sums $H_j^{(r)} = \sum_{i=1}^{j} i^{-r}$, $H_0^{(r)} = 0$. v_p is the p-adic valuation; $v_p(n!) = \sum_{k\geq 1}\lfloor n/p^k\rfloor$ (Legendre). $d_n = \operatorname{lcm}(1,\dots,n)$, so $\log d_n = \psi(n)$ is Chebyshev's ψ (all prime powers; an earlier draft wrote ϑ, which was corrected in the source notes). $\{x\} = x - \lfloor x\rfloor$. $\zeta(s) = \sum_{k\geq 1} k^{-s}$.

Part I — The first-round routes (24 September 2026)

3. The main construction and its exact algebra

For positive even n and $0 \le m \le 3n/14$ (the grid uses $m = \alpha n$, $\alpha \in \{0, \frac{1}{28}, \frac{1}{14}, \frac{3}{28}, \frac{1}{7}, \frac{5}{28}, \frac{3}{14}\}$), define

$$R_{n,m}(t) = \frac{(n!)^3}{(m!)^{14}} \frac{\left[(t-m)_m\,(t+n+1)_m\right]^7}{(t)_{n+1}^3}, \qquad L_{n,m} = \frac{1}{6!}\sum_{k\ge 1} R_{n,m}^{(6)}(k).$$

$R_{n,m}$ has poles of order ≤ 3 exactly at $0, -1, \dots, -n$, zeros of order 7 at $1, \dots, m$ and $-n-m, \dots, -n-1$, and $R_{n,m}(t) = O(t^{-2})$; hence $L_{n,m}$ converges absolutely. **[R]** Partial fractions $R_{n,m}(t) = \sum_{j=0}^{n}\left(\frac{C_j}{(t+j)^3} + \frac{D_j}{(t+j)^2} + \frac{E_j}{t+j}\right)$ have the closed form

$$C_j = (-1)^{m+j}\binom{n}{j}^3\binom{j+m}{m}^7\binom{n-j+m}{m}^7 \in \mathbb{Z},$$

with $D_j = C_j u_j$, $E_j = \frac{C_j}{2}(u_j^2 + v_j)$, where $u_j = 10H_j - 10H_{n-j} - 7H_{j+m} + 7H_{n-j+m}$ and $v_j = 10H_j^{(2)} + 10H_{n-j}^{(2)} - 7H_{j+m}^{(2)} - 7H_{n-j+m}^{(2)}$. **[R]** The coefficient of t^{-1} at infinity vanishes ($\sum_j E_j = 0$), and reflection $R_{n,m}(-n-t) = -R_{n,m}$ for even n gives $\sum_j D_j = 0$; consequently the $\zeta(7)$- and $\zeta(8)$-coefficients of the six-derivative sum die exactly and

$$L_{n,m} = A_{n,m}\zeta(9) + B_{n,m}, \qquad A_{n,m} = 28\sum_{j=0}^{n} C_j, \quad B_{n,m} = -\sum_{j=0}^{n}\left(E_j H_j^{(7)} + 7D_j H_j^{(8)} + 28 C_j H_j^{(9)}\right) \in \mathbb{Q}. \quad \text{[R]}$$

Worked example (recomputed independently for the audit, to the full precision of the archived certificate): at $(n,m) = (2,0)$, $C = (1,-8,1)$, $D = (-\frac{9}{2}, 0, \frac{9}{2})$, $E = (12,-24,12)$, $A = -168$, $B = 176.228515625$, and $L = 7.8911056302181879778007347 6895\ldots$

4. Integrality and p-adic arithmetic

Write $d = \operatorname{lcm}(1, \dots, n+m)$ and $G = \gcd_{0\le j\le n}|C_j|$. Two integerization theorems are proved in writing:

- **Elementary.** $t_{n,m}A_{n,m}$, $t_{n,m}B_{n,m} \in \mathbb{Z}$ with the raw multiplier $t_{n,m} = 2d^9/G$. **[R]**
- **Improved (headline).** For $m \le 3n/14$, $t^\sharp_{n,m}A_{n,m} \in \mathbb{Z}$ and $t^\sharp_{n,m}B_{n,m} \in \mathbb{Z}$ with $t^\sharp_{n,m} = d_n^9/G$; the two multipliers are consistent, $t_{n,m}/t^\sharp_{n,m} = 2(d/d_n)^9$. **[R, written]** The written proof removes the surface factor 2 via integral binomial Taylor coefficients and lowers the harmonic denominator bound from $n+m$ to n by a carry argument on the top prime-power level. *Caveat:* the written proof is a sketch; the carry bookkeeping is its least-protected step. Its *conclusion* (not the proof) was independently re-verified on five parameter pairs up to $n = 224$: G, d recomputed from scratch match the archived records, $t^\sharp A, t^\sharp B \in \mathbb{Z}$, and $t^\sharp/m_{\text{prim}}$ equals the archived integer. **[N, five cases]**
- **Minimality.** If $tF \in \mathbb{Z}[X]$ for a positive rational t, then $t/m_{\text{prim}} \in \mathbb{Z}_{>0}$: no clearing convention can shrink an already-computed primitive form. **[R]**

The valuations are exact: $v_p(C_j) = 3v_p(n!) + 7v_p((j+m)!) + 7v_p((n-j+m)!) - 10v_p(j!) - 10v_p((n-j)!) - 14v_p(m!)$, and $G \mid C_0 = \pm\binom{n+m}{m}^7$. **[R]** For large primes there is an exact formula: if $p^2 > n+m$ and $n \ge 2m$, then with $r = n \bmod p$, $s = m \bmod p$,

$$v_p(G) = 7 \cdot \mathbf{1}_{\,r+2s\ge 2p-1}. \qquad \text{[R; re-checked on 216 parameters / 1\{,\}951 prime terms, 0 failures, N]}$$

Small primes contribute $o(n)$, and the floor-boundary discrepancy contributes at most $7\log(n+2m+1) = o(n)$ **[R*, PNT]**. Along $m = \alpha n$,

$$\lim \frac{\log G}{n} = \Gamma(\alpha), \qquad \Gamma(\alpha) = 7\int_1^\infty \frac{\mathbf{1}_{\{x\}+2\{\alpha x\}\geq 2}}{x^2}\,dx, \qquad \lim \frac{\log t^\sharp_{n,m}}{n} = 9 - \Gamma(\alpha), \quad [\mathrm{R}^*, \mathrm{PNT}]$$

with $0 \leq \Gamma(\alpha) \leq 14\alpha$ **[R]**. Rational interval enclosures of $\Gamma(\alpha)$ (period decomposition, outward rounding on a 10^{-32} grid, explicit tail bound) give, for the six grid values, $\Gamma \in [0.14171563, 0.14171565], \ldots, [0.85210820, 0.85210822]$; all six intervals were reproduced independently. **[N]** **Boundary:** these are asymptotics of multipliers that were *proved valid*. Since $t^\sharp$ is valid, $m_{\mathrm{prim}} \mid t^\sharp$, so $9 - \Gamma(\alpha)$ is an *upper* bound for $\lim \log m_{\mathrm{prim}}/n$ — not the cost of the primitive form. The measured finite gaps $g_{n,m} = \frac{1}{n}\log(t^\sharp/m_{\mathrm{prim}})$ at $n = 224$ are $0.86, 0.71, 0.71, 0.67, 0.62, 0.74$ and show no sign of vanishing; their limit is unknown. Statements of the form "cost $\geq 9 - 14\alpha$" are lower bounds for an upper bound of the true cost and carry no consequence by themselves.

5. The contour identity and the analytic upper bound

Let $K_7(z) = \sum_{k\in\mathbb{Z}}(z-k)^{-7} = \frac{1}{6!}\frac{d^6}{dz^6}(\pi\cot\pi z)$. Two exact closed forms are used interchangeably in the corpus (both re-verified numerically for this release against a direct truncated sum, agreement to 10^{-18} at 50-digit working precision):

$$K_7(z) = \frac{\pi^7\left[302\cos(\pi z) + 57\cos(3\pi z) + \cos(5\pi z)\right]}{360\,\sin^7(\pi z)} \;=\; i\,\frac{(2\pi)^7}{720}\sum_{\ell\geq 1}\ell^6 q^\ell, \qquad q = e^{2\pi i z},\ \Im z > 0,$$

where $\sum_{\ell\geq 1}\ell^6 q^\ell = \dfrac{q\,(1 + 57q + 302q^2 + 302q^3 + 57q^4 + q^5)}{(1-q)^7}$ and the Eulerian numerator coefficients $1, 57, 302, 302, 57, 1$ sum to $720 = 6!$, which is what gives the principal part $(z-k)^{-7}$ at each integer. (In the trigonometric form the numerator coefficients $302, 57, 1$ and the denominator are both 360; the two forms are consistent. See Correction C-I, Section 8.) **[R]**

Exact contour identity. For $c \in (m, m+1)$, in particular $c = m + \frac{1}{2}$,

$$L_{n,m} = -\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} R_{n,m}(z)\,K_7(z)\,dz. \qquad [\mathrm{R}]$$

The orientation sign was re-checked independently by direct numerical contour integration at $(n,m) = (2,0)$: $I = -49.5812789531890297\,i$ and $-I/(2\pi i) = L$ to 28 digits. **[N, one case]**

Phase. With $z = nw$ and principal branches in the upper half-plane, Stirling summation gives the analytic phase

$$\Phi_\alpha(w) = -14\alpha\log\alpha + 10w\log w + 7(w{+}1{+}\alpha)\log(w{+}1{+}\alpha) - 7(w{-}\alpha)\log(w{-}\alpha) - 10(w{+}1)\log(w{+}1),$$

in which the $n\log n$ coefficient cancels exactly (factorial prefactor $3 - 14\alpha$; products $14\alpha - 3$), and

$$\log|R_{n,m}(nw)| = n\,\Re\Phi_\alpha(w) + O_\alpha(\log n)$$

at leading order — i.e. $\frac{1}{n}\log|R_{n,m}(nw)| = \Re\Phi_\alpha(w) + o(1)$. **[R; scale corrected, see C-II]** A shifted-Riemann-sum argument ($\delta = 1/(2n)$, shift of the line $\Re w = \alpha$ by δ costing $O_\alpha(\log n)$, uniformly in $y \geq 0$) upgrades this to a *uniform* estimate $\log|R_{n,m}(m + \frac{1}{2} + iny)| \leq n\Re\Phi_\alpha(\alpha + \delta + iy) + O_\alpha(\log n)$; the written argument is a sketch with untracked constants, sufficient for a limsup but not for any finite-n statement. **[M, sketch]** Combined with the contour identity, a compact Laplace bound and an exponential tail (valid because $14\alpha - 3 \leq 0$),

$$\limsup_{n\to\infty,\ m=\alpha n} \frac{\log|L_{n,m}|}{n} \;\leq\; \sup_{y\geq 0} H_\alpha(y), \qquad H_\alpha(y) = \Re\Phi_\alpha(\alpha + iy) - 2\pi y. \qquad [\mathrm{M}]$$

This is an upper bound on the exponential rate of $|L|$. The certified bounds on the tested parameter grid are positive and therefore do not establish decay. Upper bounds alone do not establish nonvanishing or a matching asymptotic formula.

The derivative has the exact form $H'_\alpha(y) = \frac{3\pi}{2} - 10\arctan\frac{y}{\alpha} - 7\arctan\frac{y}{1+2\alpha} + 10\arctan\frac{y}{1+\alpha}$, with $H'_\alpha(0) = \frac{3\pi}{2}$ and $H'_\alpha(\infty) = -2\pi$ **[R; re-derived, N-checked to 10^{-40}]**. H''_α times the positive product of the three quadratics is a quadratic in y^2 whose constant coefficient is negative, linear coefficient positive, and leading coefficient exactly $3 - 14\alpha$; hence H' has

exactly one positive zero, the global maximizer of H_α **[R, shape] [N, per α; degenerate leading coefficient at the endpoint $\alpha = 3/14$, handled by the linear root]**. Certified strict upper bounds (maximizer bracketed between exact dyadic rationals; H evaluated on the whole bracket with 256-bit Arb balls), reproduced independently to 12 digits:

α	1/28	1/14	3/28	1/7	5/28	3/14
$\sup_{y\ge 0} H_\alpha <$	0.671769718688	1.167676972547	1.609438937754	2.021873399673	2.416195251742	2.798637880122
combined bound $\lceil (9-\Gamma) + \sup H \rceil_6$	9.530055	9.885137	10.182811	10.443667	10.707140	10.946530

The combined constants add the arithmetic cost $9 - \Gamma(\alpha)$ of the proved multiplier to the real-side bound, rounded up to 6 decimals; exact rationals are archived. **All are positive: no negative exponent appears on any grid point.** They bound a limsup of a sum of two upper bounds and are not evidence about the true size of $|L|$ or $|P|$.

6. Finite certified computations (Part I) and reproduction

The first-round grid was pre-registered and executed completely, with no timeouts: 21 linear-form samples ($n = 56, 112, 224$; six positive ratios plus the $m = 0$ control) with exact rational A, B, exact d, G, primitive integer pair, and Arb enclosures certifying $|P_{n,m}(\zeta(9))| > 1$ in every case (10 positive, 11 negative); 21 Gram samples (§7); 2 regression controls. An independent re-read (re-hash, re-gcd, re-evaluation with different evaluation groupings) passed 21/21. **[N, finite]** Measured screening values $\ell_n = \log |P_{n,m}(\zeta(9))|/n$:

m/n	$n = 56$	$n = 112$	$n = 224$
0 (control)	7.301097	7.839264	7.527787
1/28	7.919132	8.527755	8.273517
1/14	8.438666	9.110458	8.657046
3/28	8.999860	9.108041	9.101122
1/7	9.117860	9.324543	9.337972
5/28	9.046996	9.909189	9.507511
3/14	9.290300	9.959672	9.790245

No profile met either pre-registered promotion gate (both larger scales <1, or strictly decreasing ℓ_n), so no $n = 448$ run was made. **These are 21 certified finite values, not evidence of a trend: along each fixed ratio the three values do not decrease strictly, and nothing here bears on infinitely many n.** Three further direct-sum certificates ($(s, n, m) = (9, 2, 0), (9, 6, 1), (9, 10, 2)$: truncated Taylor expansions of $R(k + x)$ summed in exact rationals, whole-tail control by the integral test, intervals excluding zero and overlapping the partial-fraction ζ-expression) cross-check the algebra; the $(9, 2, 0)$ case was reproduced for the audit to 28 digits. **[N]**

Reproduction. All scripts and artifacts are in the source archive (`missions/zeta9/scripts/`, `missions/zeta9/verification/`). The computations used Python 3.13 with `mpmath`, `python-flint` (Arb) and exact `Fraction` arithmetic; `verification/artifact-manifest.json` records SHA-256 hashes of scripts, notes and evidence. Displayed decimals are for presentation only; the certified objects are the archived rational numbers and Arb intervals. Ordinary floating-point values in the notes are *not* interval-certified unless explicitly said to be. The audit's own re-checks (contour orientation, phase data, Γ enclosures, sup-H bounds, improved integerization on five parameters, the large-prime formula on new data) are archived with scripts and transcripts in `verification/audit-2026-09-25/`; they support, but do not prove, the classifications above.

7. The secondary (Gram) route

Independently, a positive-weight functional for odd arguments (inherited from an earlier local study of odd zeta values) gives, at $s = 9$:

$$w_9(y) = \tfrac{2(2\pi)^8}{8!}\, y^9 \sum_{\ell\ge 1} \ell^8 e^{-2\pi\ell y}, \qquad \mu_X(t^e) = (-1)^e B_{2e+2}\tfrac{(2e+9)!}{8!(2e+2)!}, \qquad \mu_X\big((t+j^2)^{-1}\big) = j^8\big(X - H_j^{(9)}\big) - \tfrac{1}{8} + \tfrac{1}{2j}.$$

The functional and the sign of the $+1/(2j)$ term are inherited from the earlier notes (gamma integral, Euler's even-zeta formula, Hermite's formula for Hurwitz zeta); they were *not* re-derived in this effort. **[R, inherited]** With $D_a(t) = \prod_{j=1}^{a}(t+j^2)$, $h = K - N$, and $G_{ij}(X) = \mu_X\big(D_N(t)^{2q} t^{i+j}/D_K(t)\big)$, $\Delta_K(X) = \det G(X)$: at $X = \zeta(9)$ the matrix G is positive definite (an integral of a square against a positive weight), and a square Vandermonde factorization gives $\deg_X \Delta_K = h$; hence $\Delta_K(\zeta(9)) > 0$ for every K. **[R, inherited] This is the only genuine nonvanishing statement in the corpus, and it is positivity at finite K — it says nothing about size relative to the denominator.**

On the arithmetic side the corpus has a moment-integrality/rank lemma (for $p \ge 11$ the moments are p-integral for $e < 4p - 5$; at $e = 4p - 5$ von Staudt–Clausen gives valuation exactly -1; **[R*, named inputs]**) and a rank bound $\min\big(h, [K + 6N - 4p + 4]_+\big)$ adapted from the $s = 7$ pattern. **[M]** *Caveats:* the index shift was not re-derived and differs by one unit from the $s = 7$ pattern (flagged as an unverified adaptation, not as an error; no conclusion depends on it), and this is explicitly **not a determinant valuation theorem**. A structural remark records that the leading K^2 potential-theoretic terms are unchanged from $s = 7$ to $s = 9$ while the prime-by-prime integerization cost is *not* transferable. **[H]**

Finite Gram evidence: 18 records at $K = 40$ plus 3 promoted profiles at $K = 80$; all 21 values > 1; $r_K = \log P_K(\zeta(9))/K^2$ rises from $K = 40$ to $K = 80$ in all three promoted profiles ($1.343774 \to 1.429423$; $1.387844 \to 1.432339$; $1.514384 \to 1.623551$). Regression controls on the two earlier odd zeta values reproduce the archived screening values with matching coefficient hashes; one control value is the only record below one. Assignment dual certificates covering all denominator primes for the best profile at both scales give an extra cost over the primitive multiplier of ≈ 1.690479 ($K = 40$) and 1.817305 ($K = 80$) after an integer Newton change of basis; the primitive polynomial is unchanged. **[N, finite; explicitly not used as an asymptotic bound]**

8. Corrections applied in this release (v0.1)

C-I (R19 coefficient sum). The internal audit text states that the Eulerian numerator of $\sum_\ell \ell^6 q^\ell$ "sums to 360". The coefficients are $1, 57, 302, 302, 57, 1$ and sum to $720 = 6!$; 720 is exactly the value required for the principal part $(z-k)^{-7}$, and the displayed formula with the prefactor $(2\pi)^7/720$ is correct as written. The separate trigonometric form $\pi^7[302\cos\pi z + 57\cos 3\pi z + \cos 5\pi z]/(360\sin^7\pi z)$ has numerator coefficients summing to 360 against a denominator 360 and is self-consistent. Both forms were re-verified numerically against a direct sum (agreement 10^{-18} at 50-digit precision). **Impact: none on any formula, script or archived constant** — the only script using these identities uses the trigonometric form with the correct 360; the audit prose is wrong, the mathematics is not. The corrected value 720 is used throughout this note.

C-II (R20/R21 phase scale). The audit's item R20 writes the Stirling phase as "$\log|R| = \frac{1}{n}\Re\Phi_\alpha(w) + O_\alpha(1)$", inconsistent with its own R21 and with the source note. The correct scale is $\log|R_{n,m}(nw)| = n\,\Re\Phi_\alpha(w) + O_\alpha(\log n)$, equivalently $\frac{1}{n}\log|R| = \Re\Phi_\alpha(w) + o(1)$; the exact $n\log n$ cancellation ($3 - 14\alpha$ vs $14\alpha - 3$) is consistent only with the $n\,\Re\Phi$ scale. **Impact: none downstream** — the uniform estimate and the limsup bound quoted in the audit use the correct scale; only the one-line R20 statement was mis-scaled.

C-III (f_* vs τ inequality direction). With $\tau = 2641/250 = 10.564$ and f_* the maximum of the round-5–7 phase function, the certified statement (rational atanh-series bounds, independently reproduced) is

$$-10.564155 < f_* < -10.564, \qquad \text{i.e.} \qquad f_* < -\tau, \quad -f_* > \tau,$$

so the proved decay exceeds the required threshold by a margin of $-f_* - \tau = 10.5641538\ldots - 10.564 \approx 1.54 \times 10^{-4}$ per unit n. The audit's roadmap paragraph ("with $-f_* < \tau$ proved") and its gap item G8.7 ("the proved statement is $-f_* < -\tau$") both state the wrong direction; the source roadmap states $f_* < -\tau$ correctly, and the numerical margin in the audit is correct. This note uses the correct direction.

Classification hygiene enforced throughout. (i) Smallness at finitely many tested parameters, nonvanishing at finitely many tested parameters, finite certificates, and statements about infinite sequences are four different things and are never conflated here. (ii) "Not formalized" is reported as a formalization-status fact, never as a logical gap in a written proof; conversely no statement is called verified unless it is. (iii) The eight platform theorems are *abstract*; they are never cited as evidence that the *concrete* $\zeta(9)$ constructions satisfy their hypotheses. (iv) First successes inside discrete scans (e.g. the five-point cubature weights, §10) are reported as scan results, not as located thresholds.

Part II — Later rounds and the current roadmap

9. The nine-round arc

Rounds 2–9 (24 September 2026) changed the construction repeatedly. Each entry below is quoted with its boundary; the audit contains the full statements.

R	What changed / headline result	Boundary
2	Higher-order poles $p \in \{3,5,7,9\}$ with exact cofactor elimination of the other odd ζ values (seed $(w_1,w_2,w_3,w_6) = (-7776, 1701, -224, 1)$). **First exponential decay in the corpus**: the multi-ζ form $F_n = d_n^9 L_n \in \mathbb{Z} + \mathbb{Z}\zeta(3) + \mathbb{Z}\zeta(5) + \mathbb{Z}\zeta(7) + \mathbb{Z}\zeta(9)$ satisfies $0 < F_n \le e^{(-1.43+o(1))n}$. **[R]**	The form contains *four* ζ values; decay of the single-ζ(9) form after elimination is unproved (elimination-matrix rank, $A_9 \neq 0$, nonvanishing all open). 63 single-ζ(9) combinations all certified $\lvert P\rvert > 1$ **[N]**.
3	Rational shifts a/D with fixed divisor weights ($D \in \{6,8\}$); weight identities and $A = \kappa_D \rho_9 > 0$ by Lagrange interpolation; tail-branch decay $\lim \frac{1}{n}\log\lvert L\rvert = f(x_*) < -10.43$. **[R]**	No unified primitive lower bound; 31 primitive records all > 1 **[N]**.
4	**All-n nonvanishing of the leading coefficient** $\rho_n = \sum_j (-1)^j \binom{n}{j}^9 \binom{n+j}{n}\binom{2n-j}{n}$: $\operatorname{sgn}(\rho_n) = (-1)^{n/2}$ and $\rho_n \neq 0$ for *all* even $n \ge 2$, via a finite-degree real-rootedness criterion (Borcea–Brändén, Theorem 2) plus Legendre-zero and palindrome-pairing arguments. The strongest nonvanishing result in the corpus. **[R, conditional on the cited external theorem]** Also: a modular-p truncated-residue criterion for $v_p(B) = -9$ (2,428 parameter–prime checks, 4 exceptions with $v_p(B) = -8$) and a conditional obstruction ($\liminf \beta_n > 3\log 2 \Rightarrow \liminf \frac{1}{n}\log\lvert ML\rvert > 0$).	Nonvanishing of the *coefficient* ρ_n, not of the linear form. The uniform weighted density of primes with nonzero truncated residues is open; no impossibility is claimed.
5	Integer polynomial weights $W(u)$, $u = t(t+n)$, $\deg \le 4$; full-kernel basis and fixed-degree reduction; a double-layer stopping criterion ($\Gamma \le 33/4 \Rightarrow$ net cost $\ge 0.75 > 0$; $\Gamma \le 11g(3/22) = 4.58656\ldots \Rightarrow$ cost $\ge 4.41343\ldots$).	The criterion concerns *this* bound pair, not a general impossibility; no unified construction on an infinite subsequence.
6	Direction selection by the true primitive multiplier; Smith normal form of the 2×5 lattice; exact lift theorem $M(W) = 1/r$; sufficient criterion $\limsup \frac{1}{n}\log\lambda_{2,n} < 10.43 \Rightarrow \zeta(9) \notin \mathbb{Q}$; certified $r_n = -5.101469, -5.116920, -5.090410, -5.072782, -5.064944$ at $n = 12, 24, 48, 96, 192$; at $n = 192$: $\zeta(9) = a/b \Rightarrow b > 10^{422}$. **[N, finite]**	Uniformity of λ_2 is open; the criterion is sufficient, not necessary; finite r_n values certify nothing about infinitely many n.
7	Rational connection $F_{n+2} = C(n)F_n$ for all even $n \ge 2$ with exact determinant $\det F_n = \frac{(3n/2)!\,(7n/2)!}{7n\,((n/2)!)^{10}}$; $N \mid d_n^{45} d_{7n/2}^8$; sharpened real side $-10.564155 < f_* < -10.564$ (see C-III); five finite scales with weighted λ_2 upper bounds $10.50530, 10.44834, 10.47973, 10.14310, 10.27109$, all below $\tau = 10.564$. **[R, connection/det] [N, finite scales]** A **method-level obstruction**: the full-space transfer norm costs $\ge -f_*$ per unit n, so this norm route yields no strictly negative net exponent. **[R, scoped]**	The five finite crossings are finite; the obstruction constrains one norm method, not other mechanisms; no infinite-n weighted λ_2 bound exists.
8	Frozen limit model of the connection recursion: $\mathbb{Q}$-irreducibility of the characteristic polynomial (checked mod 2); no fixed block length yields a rational invariant subspace; degree-≤ 4 filters cannot give exponential cancellation; dominant eigenvalue $\rho = \exp(-2f_*) = \exp(21.128\ldots)$.	Obstructions inside a *frozen constant* model; explicitly not a no-go for the real variable-coefficient recursion. Round 8's recommendation to stop same-scale escalation is research management, not a theorem.
9	Filters of degree growing with n: for $Q \nmid P$, $\lvert P(\rho)\rvert \ge a^{-d}\left((d+1)HR^d\right)^{-4}$ with $a = 22235661$, $R = 3709731751$ (Cauchy bound); sub-linear degree, sub-exponential height filters cannot cancel.	Degree-$\Theta(n)$, height-$\exp(\Theta(n))$ filters remain conceivable if their arithmetic cost is paid; the round-7 uniform short-vector bound is still missing.

Reading. No round after round 1 produced a lower bound for $\lvert L\rvert$ or a uniform statement about a primitive denominator, and no round changed the Part-I verdict. Round 2's decay is for a form in four ζ values; round 4's nonvanishing is for a coefficient; rounds 6–7's certified values are finite; rounds 8–9 refute classes of *arguments* inside a frozen model.

10. The roadmap dependency graph (state of 25 September 2026)

After round 9 the programme was redirected into an explicit roadmap working with the $p = 9$, $m = n$, $\deg W \le 4$ family of rounds 5–7. Objects: the form matrix F_n (rows $u^0, \dots, u^4$, $u = t(t+n)$; columns (B, A_3, A_5, A_7, A_9)); the column selection S onto (B, A_9); the inverse image $E_n = SF_n^{-1}$; the saturated integer kernel basis K_n; the common denominator D_n of the primitive two-dimensional image with Smith factors $s_{1,n} \mid s_{2,n}$, $N_n = s_{2,n}/s_{1,n}$, $g_n = \gcd(N_n, d_n^2)$; weights $\mathrm{diag}(1, n^2, n^4, n^6, n^8)$; the weighted area Δ_n; the congruence-lattice first minimum μ_n; the threshold $\tau = 2641/250 = 10.564$ with the certified margin $f_* < -\tau$ (C-III). The key exact identity is

$$B_n + \sigma_n = \frac{\log(D_n/s_{1,n}) + \log \Delta_n - \log \mu_n}{n}, \qquad B_n = \frac{\log(D_n/s_{1,n}) + \frac{1}{2}\log \Delta_n - \frac{1}{2}\log g_n}{n}, \quad \sigma_n = \frac{\frac{1}{2}\log(g_n \Delta_n) - \log \mu_n}{n}.$$

Node census: 40 nodes, 13 open, 27 note-proved. "Note-proved" means a prose proof exists in a research note and was read in the audit; **none has a Lean term or a platform theorem, and the audit is not a substitute for independent verification of each one.** The 13 open nodes include the actual target J (there exist $\varepsilon > 0$ and infinitely many even n with $B_n + \sigma_n \le \tau - \varepsilon$), the stronger candidates V ($\limsup B_n < \tau$) and S ($\sigma_n \to 0$), the arithmetic budget chain VA/VB/VC/VD, the geometric sign nodes T5/T/TS/TG, the endpoint criterion FD, and the critical-scale node CP. The note-proved nodes include the corpus's **only equivalence** (CM: $\zeta(9) \notin \mathbb{Q} \iff \lambda_{2,n} = o(T_n) \iff \liminf \lambda_{2,n} Z_n = 0$, with $T_n = n^5 e^{\alpha n}$, $\alpha = -f_*$), the uniform kernel decay A ($|L_n(w)| \le e^{(-\tau + o(1))n}\sqrt{5}\|w\|_{n,2}$), the exact volume cancellation Q ($B_n = \frac{\log \Xi_n}{2n} + \frac{\log(N_n/g_n)}{2n}$), the discount identity YD, and the geometric lemmas FQ/GC/TA/FI/FC/FO/XL/TP/SP. The proved reductions are one-directional: $\mathrm{T5} \wedge \mathrm{FQ} \wedge \mathrm{X} \wedge \mathrm{A} \wedge \mathrm{F} \Rightarrow R$ (likewise with T, TS, TG); $\mathrm{CP} + \mathrm{CM} \Rightarrow R$; $V \wedge S \Rightarrow J$; $J \Rightarrow$ CP; $\mathrm{FD} + \mathrm{FC} + \mathrm{FI} \Rightarrow \mathrm{T5}$; $VA + Y \Rightarrow V$; $VB + Y \Rightarrow VA$; $VC + H + Z \Rightarrow VB$; $VD \Rightarrow VA$. Nothing proves any of the 13 open nodes; V and S are not necessary for J.

Proved negative results (with scopes). GO (a rank-2 integer polynomial lattice can meet the dimension/area/shortness data yet show both signs at three of the five points — geometry alone cannot give T5); FM (a rational connection with the same spectral limit, two-step positivity, the same five nodes and exact positive cubature can fail T5 forever — the spectral limit alone cannot give T5); the round-7 spectral obstruction and the round-8/9 filter obstructions (§9). All four refute classes of arguments; none is a no-go theorem for the route itself.

The numerical distance. V needs $\limsup B_n < \tau$; the proved split $\frac{\log \Xi_n}{2n} < \gamma = \frac{1}{4}\log(3\,711\,015\,000) \approx 5.50864282$ plus the open VA ($\limsup \frac{\log(N_n/g_n)}{2n} < \tau - \gamma \approx 5.05535718$) would give it. The tightest proved uniform estimate in that direction is $\limsup(\mathcal{E}_n^{\mathrm{mid}} - \mathcal{U}_n)/n < 22.337$, against a requirement of 0.61071437 (node VD): a gap of more than a factor of 30, in the direction where the proved bound is not small enough. Five finite parameters cross ($B_n + \sigma_n \approx 10.4933, 10.4423, 10.4767, 10.1416, 10.2703$ for $n = 12, 24, 48, 96, 192$, each below τ by 0.07–0.42); **finite crossings imply nothing**, as the roadmap itself states.

The FQ/T5 activation finding (new in the audit). The five designated samples are $k_{n,j} = \lfloor n x_* + j\sqrt{n/a} + \frac{1}{2} \rfloor$, $j = -2..2$, with $x_* = 1.00297646575489808$ and $a = 330.65816828687$ (reproduced independently to full displayed precision). Because $\sqrt{n/a} \approx 0.055\sqrt{n}$ ($1/\sqrt{a} = 0.05499337\ldots$) while $n(x_* - 1) \approx 0.00298n$, the five points become distinct and all exceed n only for $n \gtrsim 1686$ (recomputed on a discrete set of test values: degenerate at $n \le 1400$; first fully valid at $n = 2000$ *among the tested values* — no minimality over all n is claimed, and a proof for all $n \ge n_0$ would need a separate argument with controlled error and floor/rounding estimates). **Every finite parameter ever used in the effort ($n \le 384$) lies below this threshold, so no finite regression in the corpus exercises FQ or T5 at all.** Quantitatively, solving the exact five-point system for the *true* kernel moments gives weights that are not all positive at $n = 1000$ and $n = 2000$ and are positive at $n = 6000$ and $n = 20000$ (Gaussian-moment control at the same nodes is positive at every tested n and reproduces $(1/12, 1/6, 1/2, 1/6, 1/12)$ exactly; the true moment deviation decays like $n^{-1/2}$). These finite observations do not locate a threshold for positivity at all sufficiently large n; the first success in this discrete scan is not a located threshold, and the archived computations were not re-run for this release. The mechanism is sound and FQ's "eventually" remains plausible as stated; the mission's own finite certificate covers the limit constant ($330.658 < a < 330.659$), not the convergence. **[N, finite scan]**

11. The platform (formalization) layer

The local graph was partially wired into a **private** mission on a crowdsourced formalization platform (identifiers deliberately omitted from this public note). Read back from local receipts: an open root theorem ($\zeta(9) \notin \mathbb{Q}$); two accepted root decompositions (`SKETCH_ACCEPTED`, one one-form and one two-form) whose obligations require exponentially small nonzero or independent integer forms at a pinned exponential rate. **No claim of strict logical strengthening over the root is made here**; establishing such strengthening would require checking the precise coefficient, construction, and indexing constraints of the formal statements, which was not done for this release. (A child merely asserting "there exist small integer forms" would be equivalent to the root once a criterion is available — a disguised restatement, not a decomposition; the decompositions were designed to avoid at least that failure mode.) And **eight theorem-level formal proofs**, all compiled locally with Lean 4.33.1 / Mathlib and accepted by the platform: two general reductions, three abstract cores (volume cancellation, prime budget, prime discount), and three wired lemmas (a one-form irrationality criterion, a Taylor-sign lemma, an exact-cubature lemma). **Crucially, all eight have abstract statements.** The concrete objects of the roadmap (F_n, K_n, the Smith data, the weighted areas, the congruence-lattice minima) have *no Lean definitions*; consequently the accepted abstract proofs do *not* discharge the corresponding concrete local nodes, and no arithmetic arrow ($V \leftarrow VA, Q, X$, etc.) was published — deliberately, since a parent with free parameters cannot import a closed child theorem without a definition layer. None of the 27 note-proved nodes is Lean-checked. Nothing in this section is evidence that the concrete ζ(9) constructions have been formally verified.

12. Open problems (canonical closure list)

The items below include mathematical obligations for several alternative routes and, separately, formalization tasks. They are *not* asserted to be jointly necessary for an irrationality proof. A complete rigorous written proof does not require prior formalization.

12.1 Mathematical obligations

1. **C1.** An asymptotic (or matching lower bound) for the true primitive multiplier m_{prim} of the Part-I family.
2. **C2.** A lower bound for $|L_{n,m}|$, e.g. a phased saddle main term with non-cancellation (including the logarithm-branch issue in $\Phi'_\alpha(w) + 2\pi i = 0$).
3. **C3.** $A_{n,m} \neq 0$ on the Part-I range (never asserted; certified $|P| > 1$ does not imply it).
4. **C4.** The assembly $0 < |P_{n,m}(\zeta(9))| \leq e^{-cn}$ along an infinite sequence.
5. **C5.** The Gram-branch package: uniformity of the $O(K \log K)$ terms, an independently certified potential constant (the value $M = -1329/200$ used in the earlier local notes is imported and not independently certified; its literature source was not re-verified for this release), and the ζ(9) normalization cost.
6. **C7.** One of the geometric nodes T5/T/TS/TG/FD (equivalently CP at the critical scale of CM).
7. **C8.** One of the arithmetic nodes J or V∧S (hence VA, VB, VC or VD), where the proved uniform estimate still stands at 22.337 against a requirement of 0.61071437.
8. **C9.** A uniform statement over an infinite parameter set in *any* of the above — the single ingredient missing in all nine rounds.

12.2 Formalization tasks (engineering, not mathematical prerequisites)

1. **C6.** Lean definitions for F_n, K_n, the Smith data, the weighted area and the congruence-lattice minima — without them the roadmap's obligations cannot even be stated faithfully on the platform.
2. **C7†.** Formalization of the note-proved geometric layer (the formalization half of the original C7).
3. **C10.** Formal verification of the note-proved layer (27 roadmap nodes plus the Part-I proofs); today only eight abstract platform theorems exist.

13. Limitations and evidence boundaries

- **No proof in either direction.** The material cannot be combined into a proof that ζ(9) is irrational, nor that it is rational, nor that this family of constructions fails. The two structural reasons: on the main route the analytic side supplies only *upper* bounds for $|L|$ while the arithmetic side controls only *denominators* (the inequality needed points the other way, and the denominator controlled is not the primitive one); in the roadmap every closing statement is a for-all/for-infinitely-many statement while the evidence base is five-scale finite certificates. This describes the present state of these particular routes; it is *not* a claim that any route of this type must first obtain an exact asymptotic for the true multiplier or a quantitative lower bound for $|L|$.
- **Status tags are load-bearing.** **R** means "a written proof exists and no error was found on review" — review here includes re-derivation of some items and scripted re-checks of others, but not machine verification. **R*** items depend on named standard inputs (PNT via $\psi(x) \sim x$; Legendre's formula; von Staudt–Clausen; Euler's even-zeta formula; Borcea–Brändén Theorem 2). **M** items are sketches or depend on other corpus claims (R21's untracked constants; R35's unverified index shift). **N** items are finite. **H** items are not proof steps.
- **Written proofs with flagged weak points.** R8 (improved integerization) is a sketch whose conclusion — not proof — was re-verified on five parameters; R21's uniformity constants are untracked; R35's index shift is an unverified adaptation; the Gram functional (R31/R32) is inherited, not re-derived in this effort.
- **Finite evidence is finite.** All certified values ($|P| > 1$ on 21+21 samples, five-scale crossings of τ, the $b > 10^{422}$ divisor bound at $n = 192$, the cubature-weight scan) are statements about the tested parameters only.
- **Computations.** Interval-certified objects are the archived Arb balls and exact rationals; displayed decimals are presentational. Ordinary floating-point computations in the notes are not strict interval certifications. No experiment was re-run for this release beyond the small verification checks described in §8 and archived transcripts; where no run record exists, no reproduction is claimed.
- **Literature.** Only two external references are load-bearing and both are identified with checkable sources (below). The external ζ(5) construction used as a regression control in the internal notes, and the imported potential constant $M = -1329/200$, were not re-verified for this release and are not relied upon by any statement in this note. Claims from informal discussions about other odd zeta values are not used as sources.

14. AI-use disclosure

This research was conducted by the author in intensive collaboration with AI assistants (large-language-model research and coding agents): they drafted research notes, proposed constructions and proof strategies, wrote the computation scripts, ran the independent re-checks archived in `verification/audit-2026-09-25/`, and drafted the internal proof-state audit on which Part II's census is based. The author directed the programme, set the research questions, reviewed the outputs, and decided what is published. The scripted checks reported in this note are exactly the checks whose scripts and transcripts are archived — no other verification is claimed, by the author or by any tool. In particular, an AI agent's "no error found" is not independent human verification, and it is not presented as such: every such item carries the **R**/**M** tags with their stated caveats, and the closure list in §12 records what remains unproved. AI tools are not authors or contributors. This note itself was drafted with AI assistance from the archived sources and reviewed by the author; the three corrections in §8 were verified by independent numerical recomputation (scripts archived with this release).

15. References

1. W. Zudilin, *One of the numbers ζ(5), ζ(7), ζ(9), ζ(11) is irrational*, Uspekhi Mat. Nauk 56 (2001); arXiv:math/0104249. Cited for literature scoping only: the present family sits in the lineage of derivative linear forms in the central variable with parameters outside the source's stated domain, so its common-factor and saddle theorems are *not* invoked.
2. J. Borcea and P. Brändén, *Pólya–Schur master theorems for circular domains and their boundaries*, Ann. of Math. 170 (2009); arXiv:math/0607416. Theorem 2 is the external input to the round-4 all-n nonvanishing theorem (§9).

3. NIST Digital Library of Mathematical Functions, §18.2 (orthogonal-polynomial zero properties); used in round 4.
4. Internal sources (archived with this release): the nine rounds' research notes and reports (`missions/zeta9/`), the roadmap graph (`roadmap/DAG.md`, `dag.json`), the inherited odd-zeta functional notes (`missions/zeta7/research/analytic-foundations.md`), the verification artifacts and audit scripts, and the internal proof-state audit (included as provenance; its three corrected misstatements are logged in `CHANGELOG.md` and in §8 of this note).